



EE 503: Advance Communication Theory

Course Code	EE 503
Course Name	Advance Communication Theory
L-T-P-C	3-0-0-3
Prerequisites	IC-210: Probability, Statistics and Random Processes, EE-304: Communication Theory and the instructors consent
Intended for	UG/MS/PhD
Distribution	Elective for EE
Approval	5th Senate (Oct 2013)

Course Contents

- **Module I:** Review of digital modulation schemes for baseband and bandlimited channels and their corresponding optimal detectors and error probabilities [6 Lectures]
- **Carrier and Symbol Synchronization:** importance in signal demodulation, carrier frequency and phase estimation decision directed and power of N methods, timing estimation - spectral-line, MMSE, and ML methods, joint carrier and symbol synchronization. [6 Lectures]
- **Equalization:** Optimal zero-forcing equalization, Linear, Decision-feedback, Adaptive Linear, Adaptive Decision-feedback, and Blind equalization. [6 Lectures]
- **Multichannel and Multicarrier Systems:** AWGN multichannels, Multicarrier communications: OFDM modulation and demodulation, spectral characteristics, bit and power allocation, channel coding. [6 Lectures]
- **Spread Spectrum Communications:** model, Direct sequence SS, PN sequences, Frequency-hopped SS, synchronization, jamming, CDMA. [5 Lectures]
- **Introduction to Information and Coding Theories:**
 - Information Theory: information measures, Shannon entropy, differential entropy, mutual information, capacity theorem for point-to-point channels with discrete and continuous alphabets. [4 Lectures]
 - Coding Theory: linear block codes definitions, properties, bounds on minimum distance (singleton, Hamming, GV, MRRW), soft versus hard decision decoding, some specific codes (Hamming, RS, Concatenated); Convolutional codes structure, decoding (the Viterbi and BCJR algorithms); Turbo codes, LDPC codes. [9 Lectures]

References

1. T. Cover and J. Thomas, **Elements of Information Theory**, 2nd Edition, Wiley, 2006.
2. R. G. Gallager, **Principles of Digital Communication**, Cambridge Univ. Press, 2008.
3. A. Lapidoth, **A Foundation in Digital Communication**, Cambridge Univ. Press, 2009.
4. S. Lin and D. Costello, **Error Control Coding**, 2nd Edition, Prentice Hall, 2004.
5. J. G. Proakis and M. Salehi, **Digital Communications**, 5th Edition, McGraw-Hill, Prentice Hall, 2007.
6. B. Sklar, **Digital Communications: Fundamentals and Applications**, 2nd Edition, Prentice Hall, 2001.